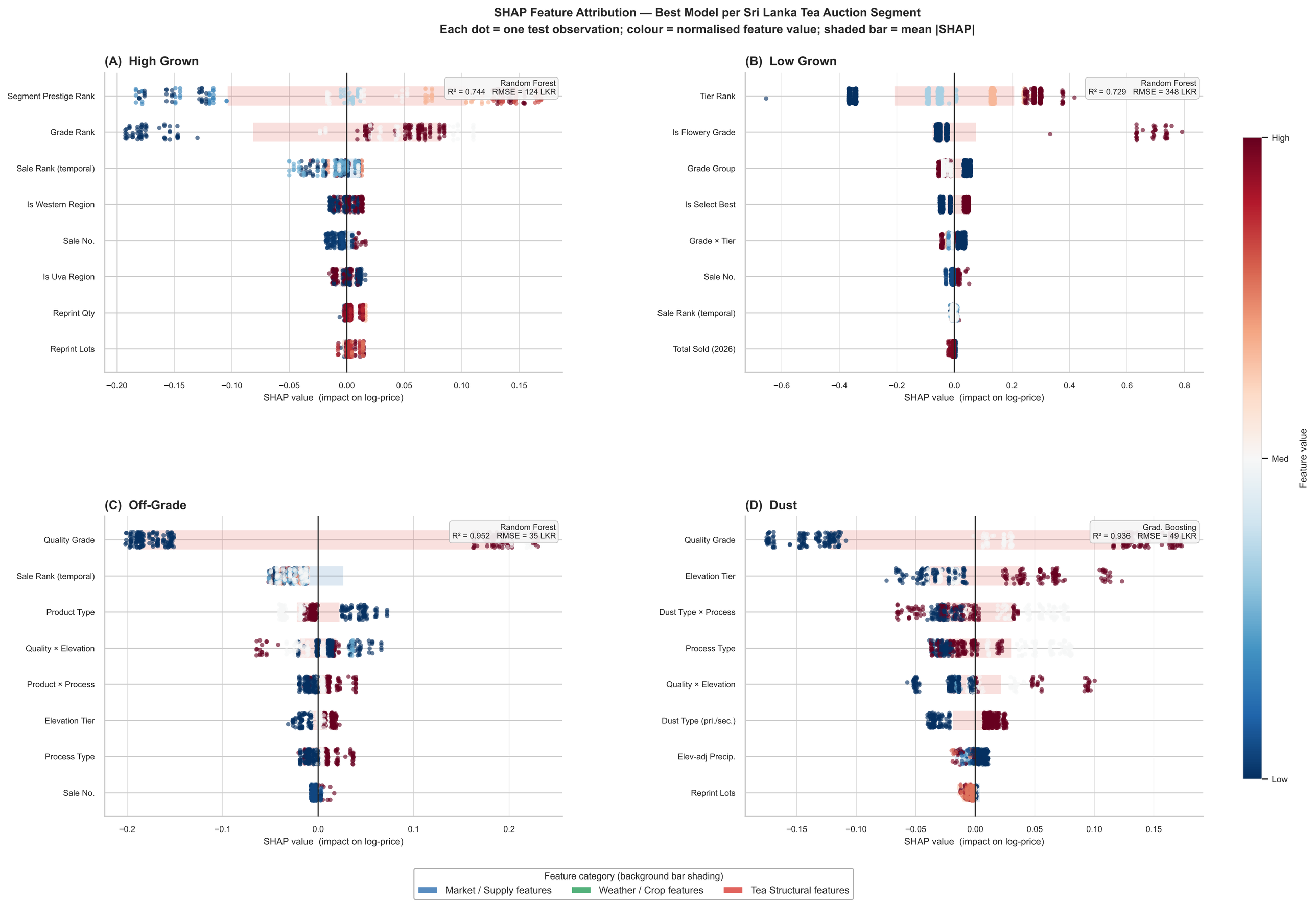
SHAP Feature Attribution — Best Model per Sri Lanka Tea Auction Segment
Each dot = one test observation; colour = normalised feature value; shaded bar = mean |SHAP|
(A) High Grown
Random Forest
R² = 0.744 RMSE = 124 LKR
Segment Prestige Rank
Grade Rank
Sale Rank (temporal)
Is Western Region
Sale No.
Is Uva Region
Reprint Qty
Reprint Lots
−0.20 −0.15 −0.10 −0.05 0.00 0.05 0.10 0.15
SHAP value (impact on log-price)
(B) Low Grown
Random Forest
R² = 0.729 RMSE = 348 LKR
Tier Rank
Is Flowery Grade
Grade Group
Is Select Best
Grade × Tier
Sale No.
Sale Rank (temporal)
Total Sold (2026)
−0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8
SHAP value (impact on log-price)
(C) Off-Grade
Random Forest
R² = 0.952 RMSE = 35 LKR
Quality Grade
Sale Rank (temporal)
Product Type
Quality × Elevation
Product × Process
Elevation Tier
Process Type
Sale No.
−0.2 −0.1 0.0 0.1 0.2
SHAP value (impact on log-price)
(D) Dust
Grad. Boosting
R² = 0.936 RMSE = 49 LKR
Quality Grade
Elevation Tier
Dust Type × Process
Process Type
Quality × Elevation
Dust Type (pri./sec.)
Elev-adj Precip.
Reprint Lots
−0.15 −0.10 −0.05 0.00 0.05 0.10 0.15
SHAP value (impact on log-price)
High
Med
Low
Feature value
Feature category (background bar shading)
Market / Supply features
Weather / Crop features
Tea Structural features